Emanuele Rossi

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Machine learning researcher developing methods to understand non-human communication, as part of a broader interest in using ML to uncover structure in complex systems: previously graph learning at Twitter and Imperial College London, and generative models for biomolecules at VantAI.

Google Scholar metrics: 4,500+ citations; h-index 15 (August 2026).

Experience

Postdoctoral Researcher, *GLADIA*, *Sapienza University of Rome* — *Oct 2025 – Present, Rome (remote)*
- Developing novel machine-learning methods to deepen our understanding of non-human communication.

Machine Learning Advisor, *Amboss Technologies* — *Oct 2023 – Present, Remote*
- Advising the ML team on graph and reinforcement learning methods for the **Bitcoin Lightning Network**.

Machine Learning Researcher, *VantAI* — *Jan 2024 – Aug 2025, NYC (remote)*
- **Co-led development of Neo-1**, a state-of-the-art all-atom latent diffusion model for multimodal structure prediction and de-novo generation of biomolecular complexes, trained on 100+ GPUs.

Machine Learning Researcher, *Twitter* — *Jun 2019 – Feb 2023, London*
- Developed **graph ML models** for Twitter's social graph and deployed a recommendation model that generated **1.2M additional daily follows**

Machine Learning Researcher, *Fabula AI* — *Mar 2019 – Jun 2019, London*
- Developed graph ML models for fake news detection as part of Fabula AI's small research team before Twitter acquired the company

Software Engineering Intern, *Google* — *Jul 2017 – Oct 2017, California*

Education

Imperial College London — *Oct 2020 – Feb 2024*
Ph.D. in Computer Science – Supervisor: Prof. Michael Bronstein
- Research focused on **graph machine learning methods** for large real-world systems

University of Cambridge — *Oct 2018 – Jun 2019*
Master in Advanced Computer Science, Distinction (**4.0 GPA**)

Imperial College London — *Oct 2015 – Jun 2018*
Bachelor in Computer Science, First Class Honors (**4.0 GPA**)

Selected Publications

Full list

Animal Communication & Audio

- *D. Marincione, D. Crisostomi, R. Dessì, E. Rodolà, and* ***E. Rossi****.* Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models. *NeurIPS 2025 AI for Animal Comm. Workshop, Spotlight (top 15%).*
- *F. Mustun, C. Semenzin, . . . ,* ***E. Rossi*** *et al.* OpenWhistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. *NeurIPS 2026, Spotlight (top 1.3%).*
- ***E. Rossi*** *and E. Rodolà.* Communicating Sound Through Natural Language. *Under review, 2026.*

Graph Learning

- ***E. Rossi*** *et al.* Temporal Graph Networks for Deep Learning on Dynamic Graphs. *ICML GRL Workshop 2020.* **1,700+ citations** and **1,200+ GitHub stars**.
- ***E. Rossi*****, F. Frasca* et al.* SIGN: Scalable Inception Graph Neural Networks. *ICML GRL Workshop 2020.* **600+ citations** and used in **production at Airbnb**.
- *B. Chamberlain, S. Shirobokov,* ***E. Rossi*** *et al.* Graph Neural Networks for Link Prediction with Subgraph Sketching. *ICLR 2023, Oral (top 5%).* **200+ citations**.

Leadership, Mentoring & Recognition

- **Leadership:** co-founded LeadTheFuture (Forbes article), a non-profit mentoring community with more than **200 mentors** and **500 selected mentees** (2018–2023).
- **Mentoring:** supervised **7 students and engineers** across Sapienza, Mila, VantAI and Amboss, resulting in **6 joint papers** (e.g. NeurIPS, LoG, IEEE ICBC).
- **Funding & recognition:** CINECA ISCRA-C compute grant (**8,500 A100 GPU-hours**, 2026); NeurIPS 2026 Top Reviewer (top 10%); Organizer, NeurIPS 2023 Temporal Graph Learning Workshop.